JOSÉ ÁNGEL SUBÍAS TOVAR

Mechatronics Engineering · Automation & Controls · Embedded Systems

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PROFILE

Mechatronics Engineering student (all coursework completed; degree expected 2027) with hands-on experience in industrial automation, embedded systems and project management, from an unattended PLC-controlled laser-cutting cell to digital tools at BMW Group. International project experience in France and Argentina.

EXPERIENCE

Underbody & Harnesses Integration Intern (incoming) *Jan 2027 – Jun 2027*

BMW Group Plant San Luis Potosí | San Luis Potosí, Mexico

- High-voltage battery integration for the BMW i3.

Project Management Steering Intern — High-Voltage Battery Plant *Jun 2026 – Present*

BMW Group Plant San Luis Potosí | San Luis Potosí, Mexico

- Support production-readiness steering: milestones, open points and cross-functional coordination.
- Developing a real-time location (RTLS) dashboard on the Sensolus API that shows live asset positions on the plant layout.
- Built internal tools: an Oracle APEX meeting-topic app and a Power Apps launch-readiness form for nine plant clusters.

Precision Manufacturing Apprentice *Jan 2026 – Jun 2026*

ITM Precision Machining Workshop | Celaya, Mexico

- CNC milling and EDM of dies, molds and parts for Tier 1/2 automotive clients, at ±0.03 mm.

PROJECTS

Automated CNC Laser-Cutting Cell — Capstone Project *2025 – 2026*

Instituto Tecnológico de Celaya

- Cell that loads material, cuts, ejects parts and restarts with no operator: Siemens S7-1214C PLC sequence and safety logic, HMI and NEMA 17 conveyor.
- ESP32 IoT layer for telemetry and remote control, with a self-hosted VPN server and a mobile app.

EASYMOTION — Soft Robotics Crutch *Sep 2024 – Jan 2025*

European Project Semester, ENIT (UTTOP) | Tarbes, France

- Student project manager and mechanical co-designer: WBS and timeline, research, SolidWorks CAD and 3D-printed prototype.

IoT Monitoring & Control Systems *Academic & personal projects*

- ESP32, Arduino, Raspberry Pi and PIC systems over MQTT and Modbus TCP, with ThingSpeak analytics and mobile apps.

EDUCATION

Instituto Tecnológico de Celaya (TecNM) — B.S. Mechatronics Engineering · GPA 89.3/100 *2021 – 2027*

ENIT (UTTOP), Tarbes, France — European Project Semester, Mechatronics & Robotics *Sep 2024 – Jan 2025*

Universidad Nacional del Nordeste, Argentina — PILA academic exchange, Information Systems *Aug – Dec 2025*

SKILLS & LANGUAGES

Automation & Controls: PLCs (Siemens TIA Portal, Allen-Bradley, Delta) · Ladder and Structured Text · HMI · servo and stepper motion · Fanuc robots

Embedded & IoT: ESP32 · Arduino · Raspberry Pi · PIC · MQTT · Modbus TCP · PROFINET · Ubuntu Linux

Software & Data: Python · MATLAB · Java · SQL (IBM Db2) · React · Oracle APEX · Power Apps · SharePoint

Design & Manufacturing: SolidWorks CAD/CAM · 3D printing · laser cutting · CNC milling and EDM

Project Management: milestone tracking · open-point management · WBS and timelines · cross-functional coordination

Languages: Spanish (native) · English (B2, TOEFL ITP 553) · French (basic, currently studying)

LEADERSHIP

- Recognized by TecNM Celaya as Student Council Representative of Mechatronics Engineering (2023–24) and Cultural Coordinator of the Mechatronics Student Association (2022).